

CSE321 Theory Assignment 02

Research and **Explore** Formal Security Models in Operating Systems by analyzing the following key areas:

- Bell-LaPadula Model
- Biba Model
- Covert Channels in Secure Systems

Report Requirements:

Write a comprehensive research-based analysis focusing on the following key factors:

Bell-LaPadula Model:

- Describe the purpose of the Bell-LaPadula model in operating system security.
- Explain how this model protects confidentiality in systems that use multiple security levels, such as Unclassified, Confidential, Secret, and Top Secret.
- Discuss the Simple Security Property, also known as “no read up.”
- Discuss the “no write down” Property.
- Analyze why Bell-LaPadula is suitable for military, government, and classified information systems, but may not be sufficient for protecting data integrity.

Biba Model:

- Describe the purpose of the Biba model and explain how it differs from the Bell-LaPadula model.
- Explain how this model protects integrity rather than confidentiality.
- Discuss the Simple Integrity Property, also known as “no read down.”
- Discuss the Integrity *-Property, also known as “no write up.”
- Analyze why Biba is useful in financial systems, medical record systems, audit logs, and other systems where unauthorized data modification must be prevented.

Covert Channels in Secure Systems:

- Define covert channels and explain how they can leak information even when formal access control rules are correctly enforced.
- Describe at least two types of covert channels, such as CPU timing channels, file-locking channels, paging-rate channels, or resource-usage channels.
- Explain why covert channels are difficult to detect and prevent in real operating systems.
- Discuss whether formal security models alone can guarantee complete operating system security.

Comparison and Critical Analysis:

- Compare Bell-LaPadula and Biba using a table.
- Clearly identify which model is confidentiality-oriented and which model is integrity-oriented.
- Discuss the practical limitations of applying strict formal security models in modern operating systems.
- Explain how formal models can be combined with practical security mechanisms such as access control, authentication, auditing, sandboxing, and mandatory access control.

Study Resources:

- Operating System Concepts, 10th Edition, Chapter 16: Security.
- Modern Operating Systems, Chapter 9: Security.
- <https://web.mit.edu/ha22286/www/papers/HST10.pdf>
- <https://www.csl.sri.com/~rushby/papers/blr86.pdf>
- <https://seclab.cs.ucdavis.edu/projects/history/papers/biba75.pdf>
- https://www.researchgate.net/publication/221367005_Towards_a_formal_model_for_security_policies_specification_and_validation_in_the_SELinux_system
- <https://www.site.uottawa.ca/~afelty/dist/mcetech17.pdf>

Submission guidelines: Can be found in the submission form given below.

Submission Link: [Submission Form](#)

Deadline: 11/5/2026, 11:00 PM

(Submissions outside the Google Form will be automatically rejected.)